Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the accuracy during training.

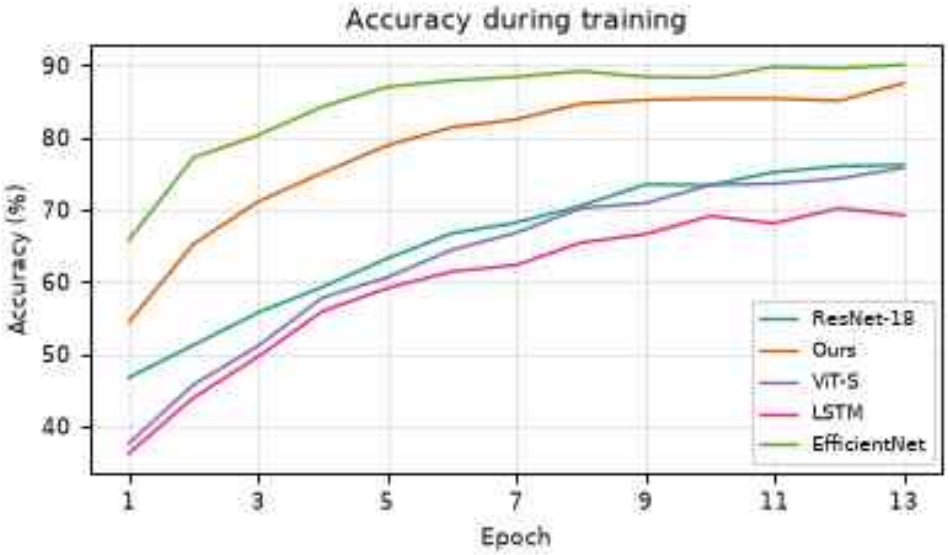


Figure 1: Accuracy during training for the models in this experiment.

From epoch 8 to epoch 10, the accuracy of Ours improves by a relative about 1%. ResNet-18 reaches 61.5% accuracy at epoch 10. ResNet-18 attains a accuracy of 56.3% at epoch 7 (Figure 1).

Figure 2 shows the f1 score by dataset.

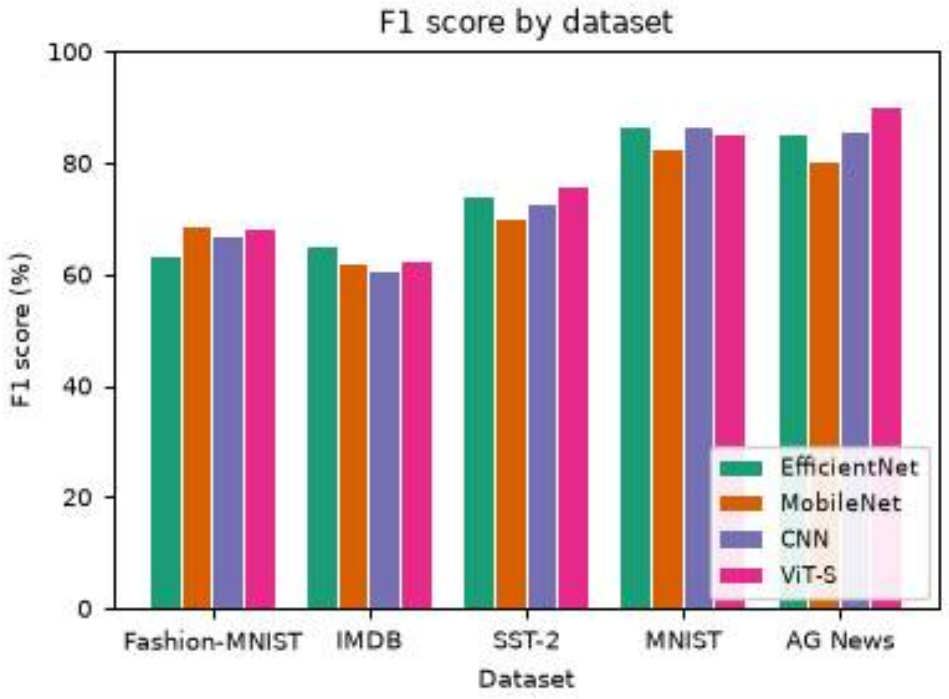


Figure 2: F1 score by dataset for the models in this experiment.

On AG News, CNN obtains a F1 score of 85% (Figure 2). As shown in Figure 2, CNN outperforms MobileNet

by 6 percentage points on AG News.

3 Discussion

Overall, the results are in line with what we expected from the literature.